Case Study: The CFV Project

CFV Corporation was one of the fastest-growing financial institutions in the country, providing customer verification, financial documentation, and credit approval services to thousands of clients every month. As the business expanded, the organization realized that its existing manual record management system could no longer support the increasing volume of customer information.

Employees were spending several hours searching for paper files, updating customer records manually, and preparing management reports. Lost documents, duplicate customer records, inconsistent data, and delayed approvals became common operational problems. Customers frequently complained about long waiting times, while managers struggled to obtain accurate and timely information for decision-making.

To modernize its operations, CFV Corporation decided to replace the manual system with an integrated **Customer File Verification (CFV) Information System**. The new system was expected to automate customer registration, electronic document management, workflow approval, reporting, user management, and communication between departments.

The management anticipated that the new information system would significantly improve productivity, reduce operating costs, enhance customer satisfaction, and support future organizational growth.

Project Procurement

After identifying its business needs, CFV Corporation prepared a **Request for Proposal (RFP)** describing the expected system functionalities. Several software companies submitted technical and financial proposals.

Following the evaluation process, **TechVision Solutions Ltd.** was awarded the contract because it submitted the lowest financial bid and promised to complete the project within twelve months. The management believed the proposal satisfied the organization's objectives and decided to proceed without conducting a comprehensive independent contract review.

Both parties signed the contract, and the software development project officially commenced.

Project Information

Item	Description
Customer	CFV Corporation
Supplier	TechVision Solutions Ltd.
Project	Customer File Verification Information System
Industry	Financial Services
Planned Duration	12 Months
Planned Budget	USD 950,000
Development Methodology	Waterfall Software Development Life Cycle
Development Team	14 Software Engineers
QA Team	3 Software Quality Assurance Engineers

Project Manager	1
Business Analysts	2
Expected Users	Approximately 500 Employees

Business Objectives

The organization expected the new software system to achieve the following objectives:

- Eliminate manual record management.
- Improve customer service delivery.
- Reduce document processing time.
- Improve the accuracy of customer records.
- Enable electronic document storage.
- Improve communication among departments.
- Generate management reports automatically.
- Increase operational efficiency.
- Reduce operational costs.
- Support future organizational expansion.

Management expected these objectives to be achieved within the agreed budget and schedule.

Initial Customer Requirements

The signed contract specified that the software should provide the following capabilities:

- Register new customers.
- Store customer records electronically.
- Retrieve customer files quickly.
- Generate operational and management reports.
- Allow multiple employees to access the system simultaneously.
- Protect confidential customer information.
- Provide a user-friendly interface.
- Reduce paperwork.
- Improve employee productivity.
- Support future expansion of the organization.

Although these requirements appeared reasonable, many were expressed in general terms without measurable acceptance criteria.

Project Development Begins

The software development team immediately began system analysis, design, and implementation according to the approved project schedule.

During the first three months, project progress appeared satisfactory. Weekly progress reports indicated that development activities were proceeding according to plan, and management believed the project would be completed on schedule.

However, as more departments became involved, numerous problems started to emerge that had not been anticipated during the proposal or contract preparation stages.

Problem 1: Ambiguous Requirements

One of the first major problems involved ambiguous customer requirements.

The contract stated that:

- The system should be **user-friendly**.
- The system should be **secure**.
- Reports should be generated **quickly**.
- Customer files should be retrieved **efficiently**.

Unfortunately, none of these statements had measurable definitions.

For example:

- What response time is considered "quick"?
- What security standard should be implemented?
- How should usability be measured?
- How many users should access the system simultaneously?

Because these questions were never answered before development began, different stakeholders interpreted the requirements differently, resulting in repeated redesigns during implementation.

Problem 2: Incomplete Requirements

As development progressed, users from different departments requested additional functionality that had never been discussed during requirements gathering.

Examples included:

- Electronic document approval.
- Role-based user permissions.
- Audit logs.
- Digital signatures.
- Email notifications.
- SMS alerts.
- Workflow automation.
- Advanced search filters.
- Backup and disaster recovery features.

Each new requirement required modifications to the existing system design, increasing both development effort and project cost.

Problem 3: Unrealistic Project Schedule

The supplier estimated the project duration primarily to remain competitive during the bidding process.

The twelve-month schedule was based on optimistic assumptions rather than historical project data or detailed workload estimation.

As development progressed:

- Software modules required more effort than expected.

- Integration activities became increasingly complex.
- Testing activities were postponed.
- Several milestones were missed.

The original project schedule gradually became impossible to maintain.

Problem 4: Poor Stakeholder Involvement

Only senior managers participated during requirements gathering.

The following groups were not consulted:

- Customer service officers.
- Records management personnel.
- IT administrators.
- Internal auditors.
- Department supervisors.
- End users.

When system demonstrations were conducted several months later, each department requested major modifications because the software did not support their daily operational activities.

This resulted in significant scope changes throughout the project.

Problem 5: Weak Risk Assessment

The proposal failed to identify several important project risks.

Examples included:

- Staff resignation.
- Delays in hardware procurement.

- Third-party software compatibility issues.
- Data migration complexity.
- Budget fluctuations.
- User resistance to organizational change.
- Cybersecurity threats.
- Network infrastructure limitations.

Since these risks had not been analyzed during contract review, the project team reacted to problems only after they occurred instead of preparing mitigation strategies beforehand.

Problem 6: Resource Constraints

Approximately halfway through the project:

- Two senior software developers resigned.
- One database administrator was reassigned.
- The project manager was assigned additional organizational responsibilities.
- Newly recruited developers required several weeks of training.

As a result, project productivity declined, communication became less effective, and software quality suffered.

Problem 7: Poor Change Management

During development, the customer continuously introduced additional requirements, including:

- Mobile application support.
- Dashboard analytics.

- Integration with external government databases.
- Electronic document sharing.
- Automatic report scheduling.

Since the project lacked a formal **Change Control Process**, developers implemented most changes without:

- Updating the project schedule.
- Revising the project budget.
- Performing impact analysis.
- Obtaining formal approval.

Consequently, the project experienced uncontrolled **scope creep**.

Problem 8: Testing Challenges

Testing activities were significantly delayed because software development exceeded planned milestones.

When testing finally began, numerous defects were discovered, including:

- Data inconsistencies.
- Security vulnerabilities.
- Performance bottlenecks.
- User interface problems.
- Integration failures.
- Incorrect business logic.
- Missing validation rules.

Many defects originated from misunderstandings that could have been prevented through effective contract review.

Project Outcome

Instead of completing the project within **12 months**, the software required **18 months** to complete.

Planned	Actual
Project Duration	12 Months
Budget	USD 950,000
Critical Defects	0 Expected
Total Reported Defects	< 50
Customer Satisfaction	High
User Training	Before Deployment
Schedule Variance	0%
Cost Overrun	0%

Although the software eventually became operational, the organization experienced significant financial losses, schedule delays, extensive software rework, and reduced confidence in the supplier.

Independent Software Quality Assurance Review

Following project completion, CFV Corporation commissioned an independent Software Quality Assurance audit.

The review concluded that most project problems originated **before software development began**.

The audit identified the following weaknesses:

- Poor requirements analysis.
- Ambiguous customer requirements.
- Inadequate stakeholder participation.
- Unrealistic cost and schedule estimation.
- Weak project risk analysis.
- Insufficient feasibility assessment.
- Inadequate resource planning.
- Lack of measurable acceptance criteria.
- Ineffective change management.
- Incomplete contract review.

The auditors concluded that a rigorous **Contract Review Process** could have prevented most of these problems before any software development activities commenced.

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Questions

Section A – Requirements Review

1. Identify at least **10 ambiguous or incomplete requirements** in the case study.
2. Rewrite each requirement using measurable acceptance criteria.
3. Which stakeholders should have participated during requirements elicitation?

Section B – Contract Review

4. Which information should have been verified before approving the proposal?
5. Which contract clauses should have been revised before signing?
6. Would you approve this contract? Justify your decision.

Section C – Risk Analysis

7. Identify at least **10 project risks**.
8. Classify each risk as:
   - Technical Risk
   - Business Risk
   - Resource Risk
   - Quality Risk
   - Schedule Risk
   - Financial Risk

Section D – Recommendations

9. Recommend at least **10 improvements** that should have been implemented before project approval.
10. Which Software Quality Assurance activities could have prevented these problems?

END